

Project Status Report

This document provides a summary of the tasks that have been fully completed in the JGH Intelligence Engine project, followed by a list of the tasks that are currently pending.

1. Completed Tasks

Model Hosting (T01)

The Qwen AI model is successfully hosted locally and made accessible through a fast backend (FastAPI).

Model Routing (T03)

An Intent Router was built to intelligently classify incoming user questions and decide how complex they are.

SQL AST Validation (T12)

The system now parses and checks generated SQL queries using sqlglot to block dangerous operations like DELETE or DROP.

Database Connection Pool (T15)

SQLAlchemy connection pooling is implemented to safely handle multiple database connections without crashing.

Schema Drift Detection (T16)

The codebase includes a schema drift detector to automatically identify if the database structure changes.

2. Pending Tasks (In Progress / Not Started)

Public HTTPS Access / Team Lead Demo (T02 & T22)

Blocked because the local server needs a tunneling service (like Ngrok or Cloudflared) to be accessible over the internet.

Schema & Business Knowledge (T04)

ChromaDB embeddings need an automated cron script to stay synchronized with business logic updates.

Multiple Users / Concurrency (T06)

Requires a Redis server to manage distributed sessions when multiple users query the AI simultaneously.

SQL Generation Accuracy (T11)

Needs improved RAG context injection so the AI generates the correct SQL based on historical examples.

Security & Privacy (T13 & T07)

Requires strict role-based permission checks and clear chat history retention logic.

Monitoring & Network Failure (T20 & T17)

Needs Prometheus/Grafana for monitoring, and a solid fallback mechanism if the local LLM crashes.

Application Upgrade & Load Testing (T18 & T21)

Requires GitHub Actions for CI/CD deployment and Locust/JMeter scripts to simulate peak user load.